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Soil & Tillage Research 63 (2002) 181–186

Soil & Tillage Research

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Short communication

Infiltration patterns into two soils under conventional and conservation tillage: influence of the spatial distribution of plant root structures and soil animal activity

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Received 7 February 2001; received in revised form 26 July 2001; accepted 10 August 2001

Abstract

The characteristics and properties of the soil macropore system may cause different infiltration behavior under different tillage practices. To evaluate the effect of a specific tillage system on infiltration and percolation with particular regard to the influence of crop structure and soil animal activity dye tracer irrigation experiments were conducted in a silty (Luvisol) as well as in a sandy loam soil (Podzolluvisol). The spatial distribution of water flow paths was experimentally examined at four square areas of 0.49 m^2, under conservation and conventional tillage. Natural rainstorms were simulated by irrigating the plots with 2.8×10^{-3} M methylene blue solutions. For both soils the root crowns of the agricultural crop, wormcasts and stained soil sections as well as macroscopic conduits were traced on plastic sheets. The investigated soil depths were 0, 5, 10 and 20 cm for the both soils. For the Luvisol, the 30, 40, 50, 80 and 120 cm depths were also studied.

For the Luvisol, the conservation tillage plot revealed pronounced vertical connectivity and continuity of the macropore network (maximum depth of stained pores = 120 cm), while at the conventional tillage plot, continuous macropores were observed to soil depths of 50 cm, but mainly restricted to the ploughed topsoil (0–30 cm soil depth).

For the Podzolluvisol, at the conservation tillage site extensive mulch residues prevented water transport beneath 5 cm soil depth. In contrast, at the conventionally tilled site stained water reached a depth of 20 cm. For all investigated plots on both soil types, the location of the root crowns of agricultural crop and of wormcasts was not related to percolation patterns.

The results suggest that conservation tillage on silty soils under agricultural landuse could induce an increased water retention capacity reducing the significance of fast runoff components.

Keywords: Infiltration; Tillage; Dye tracer; Methylene blue; Macroporosity; Preferential flow

1. Introduction

Water infiltration into agricultural soils is impacted, among other factors, by the number and connectivity of surface-vented macropores (Ehlers, 1975).

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0167-1987/02/$ – see front matter
PII: S0167-1987(01)00234-3

Characteristics and properties of the macropore network and other soil hydraulic parameters are influenced by the tillage practice (Edwards, 1982). Conventional tillage systems involve plowing and harrowing of the topsoil, while the generic term 'conservation tillage' refers to tillage systems with less soil disturbance and a higher coverage of the soil surface with mulch residues (e.g. Tebrügge and Düring, 1999).

The objective of this study is to investigate the spatial distribution, occurrence and continuity of macropores of two differently textured cultivated soils on two different sites. Both sites were subject to long-term conventional and conservative tillage systems in parallel. Additionally, the influence of soil mesofauna activity, such as earthworms, and of the position of root crowns upon the spatial distribution of macropores of cultivated soils are examined. This work is part of a study which investigates the effects of different tillage practices on water retention capacity and stormflow generation on the scale of whole watersheds.

2. Methodology

2.1. Study sites

The investigations were carried out on the two agricultural sites Adenstedt (Lower Saxony) and Lietzen (Brandenburg). The most pertinent characteristics of those sites are compiled in Table 1.

2.2. Experiments

The experiments were carried out in April/May 2000 after a dry weather period of 3 weeks. At both study sites, for each tillage method a confined irrigation plot (70 cm × 70 cm) was selected. The distance between those two plots from each other was 10 m at Adenstedt and 50 m at Lietzen, and the soil physical conditions were similar. After the superficial parts of the agricultural crop had been removed, macroscopic pore features such as cracks and holes, wormcasts and the location of the root crowns were traced on transparent plastic sheets. Subsequently, the plot was irrigated uniformly with 10 mm of a 2.8×10^{-3} M (1 g/l) methylene blue ($C_{16}H_{18}ClN_3S{\cdot}2H_2O$) solution at 30 mm/h with a portable hand sprayer. Methylene blue had been used successfully in a number of flow path visualization experiments conducted on agricultural soils (e.g. Droogers et al., 1998) and provided a favorable color contrast against the investigated soils of our study. After application of the dye tracer solution, the plots were covered with a plastic sheet to prevent evaporation. Two hours later, the staining patterns of the displaced methylene blue solution, visible unstained macropores of >1000 μm diameter (Luxmoore, 1981) and positions of the root crowns were recorded. For both, the Podzolluvisol and the Luvisol, the documented soil depths were 0, 5, 10 and 20 cm. For the Luvisol, the 30, 40, 50, 80 and 120 cm depths were also assessed. Images on the plastic sheets were digitized and processed using the GIS-software ArcView (ESRI) for stained areas, areas of unstained macropores, areas covered by root crowns, number of wormcasts, and intersections between stained areas and areas covered by root crowns.

3. Results and discussion

3.1. Staining patterns

The rate of dye tracer application proved to be sufficiently high to induce superficial water redistribution

Table 1
Site characteristics after Brunotte (1990) and Seyfarth et al. (1999)

	Adenstedt (52°00′N, 9°56′E)	Lietzen (52°28′N, 14°20′E)
Elevation	185 m a.s.l	55 m a.s.l.
Texture	Silty loam	Sandy loam
Soil type	Luvisol from loess	Podzolluvisol from glacial sands
Horizon sequence	Bt(Ap)-Bt(Bv)-(Sw)Bv_1-(Sw)Bv_2-C	Ah-Ael-Ael(Bt)-Bt-C
Soil characteristic number	75	37
Parallelly tilled since	1990	1996
Crop	Winter wheat	Rye
Tillage systems	Ploughing vs. short-harrowing	Ploughing vs. grubbering

by transitory ponding. The stained area fraction can thus be interpreted as a measure of hydraulically active macropores (Beven and Germann, 1982). At Lietzen, in 5 cm soil depth 6% of the total conventionally tilled plot surface participated in the downward displacement of the tracer solution, while at the conservation tillage site this value amounted to only 0.6% (Fig. 1a). At the conservation tillage site at Lietzen, no further downward transport of the dye tracer was observed. At the conventionally tilled site at Lietzen, the stained area decreased drastically in 10 cm and 20 cm soil depth, where tracer displacement stopped. Related field studies using methylene blue for flow path documentation report comparable area fractions participating at the tracer displacement, e.g. 2% for four clayey grassland soils (Bouma and Dekker, 1978) and between 0.4 and 2.9% for three loamy grassland soils under different management

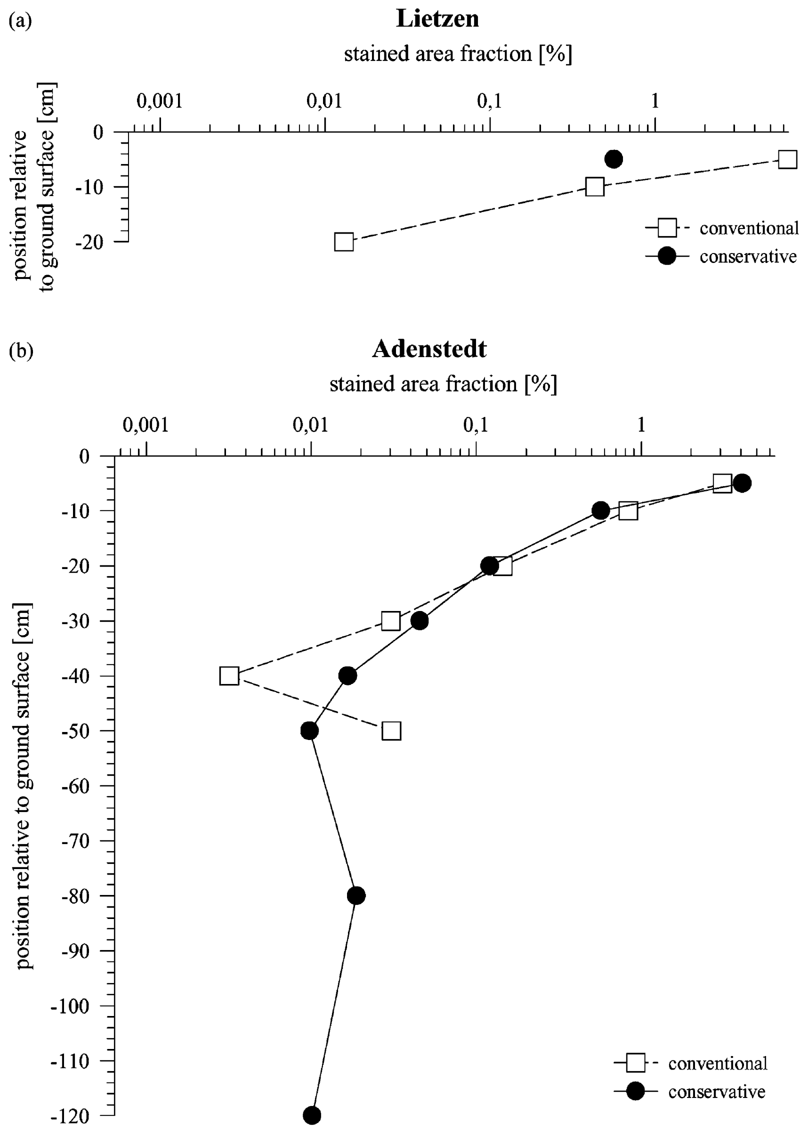

Fig. 1. 1D vertical distribution of stained area fraction over soil depth for the conventional and conservation tillage site at: (a) Lietzen; (b) Adenstedt.

(Droogers et al., 1998). Concentration of flow within certain area fractions is attributed to varying surface infiltrability as well as to the redistributing effect of the microrelief (Bouma and Dekker, 1978), which is referred to as 'distribution flow' (Ritsema and Dekker, 1995).

At the conservation tillage site at Lietzen, a mat of undecomposed plant residues in the upper 10 cm acted as a strong sorbent and probably prevented the dye tracer solution from further downward displacement (cf. Bouma and Dekker, 1978).

At Adenstedt, the area fraction of stained soil zones in 5 cm soil depth amounted to 3% at the conventional and 4% at the conservation tillage plot (Fig. 1b). Similar to the conventionally tilled plot at Lietzen, the stained area sharply decreased with depth. Beneath the intensively rooted soil zone at about 10 cm soil depth, the occurrence of dye tracer is essentially restricted to macropores. While the greatest depth of observed staining at the Adenstedt site was 50 cm at the conventionally tilled plot, stained macropores were found down to a soil depth of 120 cm at the conservation tillage plot. This implies a greater continuity of the macropore network at the conservation tillage plot and is in accordance with findings in different loess soils (Ehlers, 1975; Beisecker, 1994). A greater pore continuity and connectivity in soils under conservation tillage has been attributed to higher activity of bioturbative soil animals, such as earthworms (Beisecker, 1994).

3.2. Earthworm activity

At Adenstedt, the number of wormcasts per square meter amounts to 57 for conventional tillage and 69 for conservation tillage, respectively. This is comparable to the 55 wormcasts per square meter reported by Ehlers (1975) for a silt loam under no-till. The difference in the number of wormcasts between conventional and conservation tillage indicates a greater earthworm activity under conservation tillage. This is in line with the findings of Joschko et al. (1997), who report for the Adenstedt site higher earthworm abundance on conservation tillage as compared to conventional tillage.

Greater earthworm abundance under conservation tillage as compared to conventional tillage is also reported for the Lietzen site (Seyfarth et al., 1999). However, this is in contrast to our results, since at the Lietzen site no wormcasts were found at the conservation tillage plot. Possibly the complete absence of wormcasts at the surface of the conservation tillage plot is due to alternative cast deposition opportunities along an extended macropore system, while no such opportunities exist at the conventionally tilled plots, the pore system of which is basically limited to the soil section above the plough layer. Another possible reason could be a masking of wormcasts by plant residues. A high spatial variability of soil physical and biological properties within uniformly tilled sections at Lietzen (Seyfarth et al., 1999) may also account for this unexpected finding.

3.3. Unstained macropores

In the sandy loamy Podzolluvisol at Lietzen unstained macropores occurred predominantly as fine linear cracks, the total area fraction of which could not be assessed. Therefore, further reflections on areal coverage of unstained macropores refer to the two plots at the silty loamy Luvisol at Adenstedt. There the areal coverage (Fig. 2) exhibits a minimum at 50 cm depth for both, the conventional and conservation tillage plot. This depth approximately matches the compacted zone below the plow pan, which extends from 30 to 50 cm depth. At the surface of the conventionally tilled plot, the areal coverage of unstained macropores amounts to 0.7% (Fig. 2). Similar values of 0.1–2% were obtained for macropores >400 μm of fine-loamy and fine-silty soils under different tillage treatments by e.g. Logsdon et al. (1990). The relatively high 2–12% areal coverage of macropores >60 μm determined at four clayey soils by Bullock and Thomasson (1979) seem to be related to clay-specific crack development. At Adenstedt, the lower areal percentage covered by macropores at depths $\geq$20 cm of the conventionally tilled plot as compared with the conservatively tilled plot (Fig. 2) may be due to a tillage-induced disturbance of the topsoil, causing adverse effects on the activity of bioturbative soil animals. In contrast, macropores at the ground surface as documented before tracer application show higher values at the conventionally tilled site compared with the conservatively tilled plot and might be caused by mechanical loosening in the course of tillage operations (Lindstrom et al., 1984).

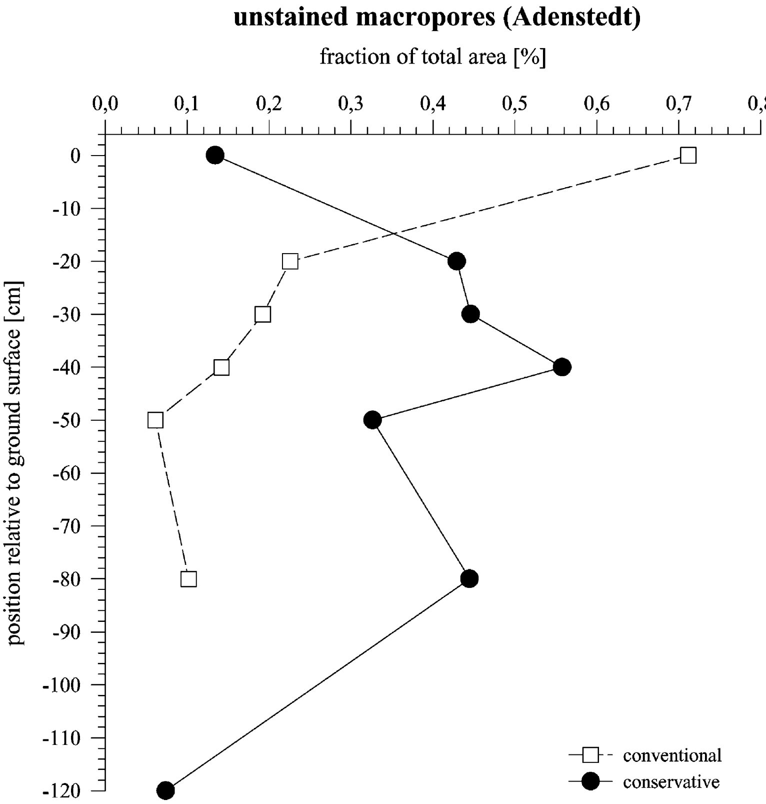

Fig. 2. 1D vertical distribution of unstained macropores over soil depth for the conventional and conservation tillage site at Adenstedt.

Besides a deeper penetration of dye tracer at the conservation tillage site (Fig. 1b), the different areal fractions of unstained macropores of the two tillage systems at the Adenstedt site (Fig. 2) are not reflected by the stained area fractions (Fig. 1b). For both tillage treatments, the total stained area fraction in 5 cm soil depth is much higher than that of the visually recognizable unstained macropores at the surface (Figs. 1b and 2), where all macropores, regardless of staining, were recorded. The high staining percentage at 5 cm soil depth may reflect macropore flow along horizontally oriented cracks or interpedal surfaces, or, more probably, a significant proportion of flow through meso- and micropores.

3.4. Crop structure

The funneling effect of the crop stems and root-channels may play an important role upon water infiltration into soils (e.g. Mitchell et al., 1991). At Adenstedt and Lietzen, however, no such effects were observed, since the positions of root crowns and staining patterns in 5 cm soil depth were not correlated. Here, rather the linear microtopography formed by the ridges and the depressions between the crop rows comprising an altitudinal difference of about 5 cm could have induced the horizontal 'distribution flow' (Ritsema and Dekker, 1995) of the applied dye tracer solution on the soil surface.

4. Conclusions

Under a conservation tillage system deep percolation in silty soils is probably caused by a highly continuous macropore network that often is missing under a conventional tillage system. On the other hand, water displacement under a conservation tillage system can be diminished by sorptive plant residues incorporated in the topsoil. An increased macroporosity under conservation tillage due to favorable conditions for burrowing soil animals, such as earthworms, was reflected by a higher number of wormcasts for only one of the two study sites. Consequently, without supplementary data, wormcasts are an unreliable measure for bioturbative activity. A distributing effect of crop roots upon percolation patterns in the topsoil was not found in our study, irrespective of tillage practice or soil texture at the four investigated plots.

Due to its high potential for increased water infiltration and displacement to greater soil depths, conservation tillage could increase the water storage capacity of cultivated areas. In this respect, conservation tillage could be envisaged as a factor in water retention and reduction of small stormflows.

Acknowledgements

This investigation was financially supported by the German Ministry for Education and Research (BMBF) as subproject A 4.3 of the German Research Network Natural Disasters (DFNK). For putting the experimental site Lietzen at our disposal the authors are grateful to Dr. M. Joschko (ZALF e.V., Müncheberg). Assistance in editing the digitized structures by E. Gassert, Chair of Hydrology and Water Management (BTU Cottbus), is highly appreciated.

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